ASHMILA ASHRAF

SOFTWARE ENGINEER

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PROFESSIONAL SUMMARY

Software Engineer with 2.7 years of experience developing secure web applications for national-level government systems. Proficient in end-to-end development using React.js, Spring Boot, and Node.js, with a strong focus on application security and real-time monitoring systems. Experienced in vulnerability assessment and remediation aligned with OWASP WSTG standards, along with web application security testing, API integration, and secure authentication implementation. Proficient in Python for problem-solving.

EXPERIENCE

Project Engineer **Jun 2023 - Feb 2026**
Centre for Development of Advanced Computing (CDAC) *Thiruvananthapuram, KL, IN*

- Independently architect, develop, and deploy **Full-stack web applications** for national projects, managing complete lifecycle from requirements to production deployment
- **Application Security:** Remediated critical vulnerabilities post internal penetration testing includes authentication & authorization, credential encryption, server-side session management, IP-based rate limiting with account lockout, error handling, security headers configuration, input validation & sanitization, and XSS/SQL injection prevention
- Developed secure authentication using Keycloak SSO with HttpOnly cookies, RSA/ AES encryption, API-key authentication, RBAC and environment-based secrets management
- Build RESTful APIs for data management, real-time updates, and frontend integration; integrate multiple internal department APIs for complex data visualization with geospatial mapping (React Leaflet)
- Managed application deployment; developed shell scripts for deployments; configured NGINX, WildFly, and Syslog-ng on RHEL 9.4
- **VAPT :** Application/API risk assessment, secure code reviews, and web application security testing using Burp Suite, Snyk (SAST/SCA), Postman and manual testing methodologies per OWASP WSTG standards

PROJECTS

ERSS Scenario Replay (CDAC) **Jul 2023 - Sept 2024**

- ***Technology:*** *ReactJS 18, NodeJS, React Chrono, React Leaflet, Material UI, Keycloak, NGINX*
- Developed comprehensive timeline visualization application for end-to-end emergency incident data representation with multimedia integration, nested timelines for multi-vehicle dispatch tracking, geospatial mapping with vehicle route visualization and PDF report generation
- Implemented secure authentication - AES & Keycloak SSO
- Deployed across states - Phase 1 & 2 of **Emergency Response Support System (ERSS)**

ERSS Alarms and Alerts (CDAC) **May 2024 - Oct 2025**

- ***Technology:*** *ReactJS 17, Spring Boot 2.6, Java 11, Python 3, PostgreSQL, Syslog-ng, Wildfly 21*
- Transformed partially developed system into production-ready, multi-state real-time log monitoring platform with centralized national dashboard featuring interactive India map.
- Built full-stack solution: ReactJS frontend, Spring Boot backend with RESTful APIs, log capturing using Syslog-ng and Python scripts for log processing.
- Centralized Logging System: Set-up Syslog-ng server configuration for capturing logs from diverse technology stack (Java, Node.js, Python applications).
- Implemented comprehensive security hardening post-audit.

ERSS Call Center Performance Monitoring System (CDAC) **Mar 2025 - Dec 2025**

- ***Technologies:*** *ReactJS 19 (Vite), NodeJS, Keycloak, Material-UI, Radix UI, Shadcn UI, Lucid react*
- Developed a real-time performance and availability monitoring web application for ERSS call taker centers, featuring live agent tracking, threshold-based alerts, real-time status indicators, and interactive analytics dashboards.
- Built comprehensive report pages with filtering and export capabilities; integrated REST APIs from internal reporting team and implemented secure role-based authentication using Keycloak.

Additional Contributions

- Maintained and supported ERSS Call Tracing application, handling bug resolution, issue troubleshooting, and deployments.
- Designed reusable Syslog-ng configuration templates for Java, Node.js, and Python applications, adopted across multiple development teams for standardized logging.
- Authored technical documentation including system architecture, API specifications, and deployment procedures to support development and onboarding

ACADEMIC PROJECT

SignQuiz **2020**

- ***Technology:*** *Python, Django, TensorFlow, Keras, OpenCV*
- Developed a web-based quiz application for learning Indian Sign Language, integrating basic image recognition capabilities.
- Worked on implementing webcam-based image capture and integrating a pre-trained CNN model for sign recognition.

SKILLS

- **Programming Languages:** Python, JavaScript
- **Frontend Technologies:** ReactJS (v17, v18, v19 - Vite), HTML5, CSS3
- **Backend Frameworks:** Node.js, Spring Boot
- **React Ecosystem:** Material-UI, React Bootstrap, React Chrono, React Leaflet, React Redux
- **Database:** PostgreSQL
- **Security Practices / Management:** Secure Code Review, Risk Assessment, Keycloak, API-key
- **Security & Testing Tools:** Burp Suite, Snyk, Postman, Bandit
- **API Development:** RESTful APIs, API Integration
- **DevOps / Tools & Platforms:** Git, Github, Nginx, WildFly, RHEL, PgAdmin, Syslog-ng, AWS, Render, Vercel, Neon
- **AI Tools:** Claude ai, v0 by Vercel, Lovable, Chatgpt, Anti-gravity, Gemini, Deepseek
- **Other:** Problem Solving, Manual Debugging, Security Vulnerability Assessment & Remediation

ACHIEVEMENTS

- **Copyright Registration:** ERSS Scenario Replay web application Registered under the Indian Copyright Act, 1957 (Certificate No. SW-2025020784)

EDUCATION

Bachelor of Technology in Computer Science and Engineering **2016-2020**
Malabar Institute of Technology - APJ Abdul Kalam Technological University
CGPA: 8.79

LANGUAGES

- English, Malayalam